Project Initialisation and Planning Phase

Date	12 July 2026
Team ID	
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal

Project Overview	
Objective	To help students receive fast, emotionally-aware academic support by automatically detecting their emotional state from free text and generating personalized, empathetic guidance in real time.
Scope	Covers text-based emotion classification (BiLSTM + BERT), Gemini-powered response generation, CSV-based interaction logging, and a Streamlit analytics dashboard for visualizing emotion and model-confidence trends, for self-paced learners and academic support teams.
Problem Statement	
Description	Students often struggle silently with confusion, frustration, or boredom while studying independently, and manually reading tone or emotion from text-based questions is slow and inconsistent for instructors and TAs, delaying the right kind of support.
Impact	Solving this improves how quickly and empathetically students receive help, reduces manual triage effort for academic staff, and gives instructors visibility into class-wide emotional trends through analytics.
Proposed Solution	
Approach	Combine deep-learning emotion classifiers (BiLSTM and BERT) with Gemini-generated natural-language guidance, so free-text input is automatically classified into one of five emotional states and answered with tailored, empathetic tips.
Key Features	Dual-model (BiLSTM + BERT) emotion classification with confidence scoring and mixed-emotion detection Gemini AI-generated personalized guidance, with an offline fallback handler for reliability CSV-based interaction logging for every submission Streamlit analytics dashboard (Plotly) comparing model confidence and emotion distribution over time

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	Kaggle / Colab T4 GPU (for BiLSTM & BERT training)
Memory	RAM specifications	16 GB (recommended for BERT inference)
Storage	Disk space for data, models, and logs	10 GB SSD (model artifacts + CSV interaction logs)
Software		
Frameworks	Python frameworks	Streamlit, TensorFlow/Keras, PyTorch
Libraries	Additional libraries	transformers, numpy, pandas, plotly, scikit-learn, nltk
Development Environment	IDE	Jupyter Notebook, VS Code (Python 3.11 venv)
Data	Source, size, format	Emotion-labelled student text dataset (custom-curated + public emotion corpora), 5 classes (Confused, Frustrated, Curious, Confident, Bored), CSV format